

Introduction to Intelligent Robotics (CC3046)

Project Proposal

RoboCup Two-Robot Skills: Comparing Rule-Based,
Evolutionary and Hierarchical Reinforcement Learning
Strategies

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April 2024

Summary of the goals of the work

While deep reinforcement learning has demonstrated overachieving success in various domains, its complexity and computational demands often pose challenges, particularly in resource-constrained environments such as robotic soccer simulated environments. Thus, this project aims to explore simpler, traditional reinforcement learning and other machine learning algorithms, along with rule-based decision making strategies, within the available environments in the RoboCup simulator.

In particular, the objectives defined for this work focus on the equipment of robotic soccer agents with coordination skill sets for effective two-player gameplay, such as passing and kicking, and executing effective corner kicks, by employing the aforementioned traditional reinforcement and machine learning techniques. Hence, we seek to achieve the following outcomes:

- Development of an effective coordination and strategy system for two-robot teams that does not rely on deep reinforcement learning.
- Optimisation of a set of skills and manoeuvres for robotic soccer, applicable in various game scenarios.
- Implementation of communication protocols to ensure effective and efficient information sharing among agents.
- Evaluation and comparison of the effectiveness of the learnt skills derived from the various training and algorithmic adopted procedures.

Proposed approach to solve the problem

Description of the setup

This project will be fully developed on the robotic soccer environment simulator *RoboCup*. The simulator provides a virtual environment where robotic soccer agents can interact with each other and the ball. It leverages a physics engine to simulate the dynamics of the soccer field, ball and robot movements accurately. Moreover, it can be incorporated with a visualisation tool to allow the observation of the robotic agents in the simulated field. In this case, the visualisation tool is Offenburg University's *RoboViz*.

The robots in the simulated environment are equipped with a range of sensors and actuators, to perceive and act upon the world. These include joint actuators for controlling various body parts, such as legs, arms, knees, elbows and head. Additionally, sensors provide information of the robot's localization, orientation, joint angles, and the position of surrounding objects, for example, the ball, goalposts, and other players.

The simulated environment encapsulates various elements crucial for gameplay. It includes a soccer field with defined dimensions and boundary lines. Additionally, it incorporates static obstacles like goalposts and outer field lines, along with dynamic obstacles such as other players and the ball. The environment parameters also define constraints on robot movement, such as maximum walking speeds and joint angle ranges. Furthermore, the environment facilitates communication between agents, enabling teamwork through features like radio communication for sharing positional information.

Algorithms

The chosen, proposed algorithms and techniques for this work are Rule-based Decision Making, Evolutionary Strategies, and Hierarchical Reinforcement Learning, within the RoboCup environment, to study and compare their performances in the context of learning two-agent skills coordination. A brief description of each approach is presented below. This selection rises from their shared characteristics of being model-free. Moreover, these algorithms offer simplicity and transparency, presenting advantages over more complex deep learning approaches.

Rule-based Decision Making

Rule-based Decision Making will involve the creation and execution of predefined rules or heuristics to guide the behaviour of the soccer robot agents. For example, rules may dictate when to pass the ball, or when to move into a scoring position. However, this approach does not involve any learning processes. Therefore, they serve as a valuable baseline for comparison against the following, more complex learning algorithms.

Evolutionary Strategies

Evolutionary Strategies offer a population-based optimization approach for learning two-agent skills coordination. It operates by iteratively evolving a population of candidate solutions, each representing a potential strategy or behaviour for the agents. Through a process of mutation and selection, the algorithm explores the space of possible strategies, favouring those that demonstrate improved

performance in achieving the desired coordination tasks, such as passing and scoring goals. Unlike traditional reinforcement learning methods, it does not rely on explicit gradient-based optimization or value function approximation.

Hierarchical Reinforcement Learning

Hierarchical Reinforcement Learning (HRL) involves organising the learning process into multiple levels of abstraction, with each level representing different temporal or spatial scales of decision-making. At the top level, a higher-level policy or controller determines high-level strategies or goals, such as overall game tactics or team formations. At lower levels, lower-level policies or controllers handle finer-grained actions and interactions, such as individual player movements or ball passing. HRL allows agents to learn complex coordination skills efficiently by focusing on both global strategies and individual actions.

Performance Evaluation Experiments

Experiment 1 – Passing and Kicking

- **Experimental Setup:**
 - Agents execute passes and kicks to predefined targets, such as teammates or goalposts.
 - 100 runs are conducted for each approach to ensure robustness and reliability of results.
- **Metrics:**
 - **Success Rate:** Percentage of successful passes and kicks out of total attempts.
 - **Distance from Target:** Measure of accuracy, indicating how close the ball reaches the intended target.
 - **Time of Execution:** Duration taken by agents to execute each pass and kick.

Experiment 2 – Corner Kicks

- **Experimental Setup:**
 - Conduct experiments to evaluate the effectiveness of corner kicks in creating scoring opportunities.
 - Perform 100 corner kicks for each approach to gather sufficient data.
- **Metrics:**
 - **Goals Scored:** Number of goals resulting directly from corner kicks.
 - **Passes Completed:** Number of successful passes made by the receiving player/team after the corner kick.
 - **Average Setup Time:** Mean duration taken to set up and execute each corner kick.

Milestones Outline

Weeks Leading up to the Checkpoint (Weeks 1-3)

Week 1 (23/04 - 30/04): Project Initialization, Simulation Setup and Paper Outline

1. Set up the RoboCup simulation environment and familiarise with Offenburg University's RoboViz visualisation tool.
2. Conduct preliminary tests to ensure the simulation environment is working as intended.
3. Begin drafting an outline for the scientific paper, focusing on the introduction, which should contextualise the use of traditional reinforcement learning and other machine learning techniques in robotic soccer.
4. Establish initial communication protocols for basic agent interaction within the simulation.

Week 2 (30/04 - 07/05): ES and HRL Implementation and Initial Experiments

5. Implement initial versions of Evolutionary Strategies and Hierarchical Reinforcement Learning within the simulation.
6. Develop and test basic agent coordination skills like passing and kicking.
7. Start writing the methodology section of the paper, detailing the simulation setup, algorithms used, and the initial experiments designed.
8. Execute preliminary experiments to test the basic functionality and coordination of the agents.

Week 3 (07/05 - 14/05): Refinement and Paper Progression

9. Refine the implemented algorithms based on the outcomes of initial tests.
10. Enhance the agent coordination skills, preparing for more complex tasks like corner kicks.
11. Update the scientific paper with early results and refine the methodology section based on the progress.
12. Prepare for the first round of comprehensive experiments focusing on passing and kicking.

Objectives for Intermediate Delivery (Checkpoint) - 14/05

- Successfully implemented Evolutionary Strategies and Hierarchical Reinforcement Learning in the simulation.
- Demonstrated basic agent coordination skills (passing and kicking) in the simulation environment.
- Initial draft of the scientific paper with completed introduction, methodology, and preliminary results sections.
- Execution plan for the first set of performance evaluation experiments.

Weeks Leading up to the Final Delivery (Weeks 4-6)

Week 4 (14/05 - 21/05): Rule-Based Decision Implementation, and Advanced Experiments and Analysis

13. Implement Rule-Based Decision Making within the simulation.
14. Conduct the Experiment 1 – Passing and Kicking, collecting data on success rates, accuracy, and execution times.
15. Begin analysing the collected data to identify patterns, strengths, and weaknesses of each algorithm.
16. Update the scientific paper with the results of the first experiment and start discussing the findings.

Week 5 (21/05 - 28/05): Further Experiments and In-Depth Analysis

17. Execute Experiment 2 – Corner Kicks, focusing on goals scored, passes completed, and setup times.
18. Analyse the results, comparing the effectiveness of each approach in the context of corner kicks.
19. Continue updating the scientific paper, incorporating detailed analyses and discussions on both sets of experiments.

Week 6 (28/05 - 03/06): Finalization and Presentation Preparation

20. Finalise all experimental analyses and consolidate findings, drawing conclusions about the effectiveness of the different algorithms in robotic soccer coordination.
21. Complete the scientific paper, adding the conclusion section and ensuring all data and discussions are clearly presented and supported by the results.
22. Prepare a comprehensive presentation that outlines the project's objectives, methodologies, key findings, and implications of the research.

Objectives for Final Delivery (Checkpoint) - 03/06

- Completion of all planned experiments with a thorough analysis of the results.
- A scientific paper detailing the project's approach, experiments, results, and conclusions, ready for submission or presentation.
- A detailed presentation showcasing the project's journey, findings, and the potential impact of the research in the field of robotic soccer and machine learning.